Phishing Email Detection & Awareness System

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SUMMARY.

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Introduction.

Phishing is a cyberattack technique used by cybercriminals to deceive users into revealing sensitive information such as passwords, banking details, or personal data. These attacks are typically conducted through emails that imitate legitimate organizations.

The objective of this project is to analyze several emails in order to identify phishing indicators and classify messages according to their level of risk.

Methodology.

The analysis process used in this project includes several steps.

EMAIL COLLECTION

A dataset of emails was collected, including both phishing messages and legitimate emails. These samples serve as the dataset for analysis.

EMAIL HEADER ANALYSIS

Email headers contain technical information such as:

- sender IP address
- SMTP server used
- message transmission path

These details help identify the real origin of the message.

DOMAIN AND LINK INSPECTION

Links contained in emails are examined to detect:

- suspicious domains
- recently registered domains
- URLs imitating legitimate websites

Tools such as:

- Toolbox
- MXToolbox

can be used to analyze these links.

IDENTIFICATION OF PHISHING INDICATORS

Several indicators may signal a phishing attempt, including:

- urgent or alarming language
- spelling or grammar errors
- requests for sensitive information
- suspicious links

RISK CLASSIFICATION

Emails are classified according to three levels:

Low : Legitimate email
Medium: Suspicious email
High : Confirmed phishing

Email Analysis.

SIMPLE ANALYSIS

EMAIL ID	TYPE	INDICATORS	RISK LEVEL
01	Phishing	Suspicious link + urgency	HIGH
02	Phishing	Request for banking information	HIGH
03	Phishing	Fake delivery notification	MEDIUM
04	Phishing	Security alert message	HIGH
05	Phishing	Fake shared document	MEDIUM

EMAIL ID	TYPE	INDICATORS	RISK LEVEL
06	Legitimate	Internal communication	LOW
07	Legitimate	Project information	LOW
08	Legitimate	Interview invitation	LOW
09	Legitimate	IT maintenance notice	LOW
10	Legitimate	Internal training reminder	LOW

The results show that phishing emails often contain external links and use social engineering techniques to encourage the user to act quickly.

HEADERS ANALYSIS

SAMPLE	SENDER	SENDER DOMAIN	SPF Status	DKIM Status	DMARC Status	MessageID	Risk Level
1	BANCO DO BRADESCO LIVELO banco.bradesco@atendimento.com.br	atendimento.com.br	temperror	none	temperror	20230919183549.39DEA3F725@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06	**High Risk**: Temporary SPF/DMARC error, suspicious MessageID (Ubuntu server). Likely malicious or spoofed email.
2	Microsoft account team no-reply@access-accesecurity.com	access-accesecurity.com	none	none	permerror	032672b4-77ca-42f8-a036-9711e91bd1f3@DB8EUR06FT032.eop-eur06.prod.protection.outlook.com	**High Risk**: Sender domain is deceptive (typosquatting "accesssecurity"). All authentication checks failed. Classic phishing attempt.
3	Zonnepanelen installateur zonnepaneel@appj.serenitepure.fr	appj.serenitepure.fr	none	none	none	0.0.0.0.1D8EF409A5C12CE.37AA@dturm.de	**High Risk**: No authentication methods passed. Sender domain does not match the subject. Very suspicious.

SAMPLE	SENDER	SENDER DOMAIN	SPF Status	DKIM Status	DMARC Status	MessageID	Risk Level
4	BANCO DO BRADESCO LIVELO banco.bradesco@atendimento.com.br	gmail.com	**pass** (IP Unknown)	**pass** (domain Unknown)	**pass**	fe5a59263be048e994229d8d1b31df82@gmail.com	**Suspicious / Moderate Risk**: Authentications technically pass, but it's a personal Gmail address. The subject ("IRPF Release") is typical of financial scams. The sender account is likely legitimate but compromised.
5	Microsoft account team no-reply@access-accessecurity.com	access-accessecurity.com	none	none	permerror	412928a2-0aad-4b27-959a-5df404e1f07d@V11EUR06FT066.eop-eur06.prod.protection.outlook.com	**High Risk**: Similar to sample 2. Fraudulent domain attempting to impersonate Microsoft. Authentication missing or error

- **Samples 1, 2, 3,** and **5** show strong signals of phishing or identity spoofing. Failures or errors in authentication mechanisms (SPF, DKIM, DMARC) indicate that the sending server is not authorized by the owner of the domain used in the "From" field.

- **Sample 4** is more ambiguous. The headers are technically valid (pass), meaning Gmail's servers did indeed send this message. This most likely indicates a legitimate Gmail account that has been compromised and is being used to send malware or phishing campaigns.

- **Key Observation: Samples 2** and **5** use domain names that are very close to legitimate names ("access-accesecurity.com" instead of "accesssecurity.com" or "microsoft.com"). This is a common typosquatting technique used in phishing.

Discussion.

The analysis showed that phishing attacks mainly rely on:

- impersonation of well-known organizations
- creating a sense of urgency
- redirecting users to fake websites

These techniques exploit the human factor rather than technical vulnerabilities.
Organizations should therefore combine:

- technical security solutions
- user awareness programs.

Prevention Recommendations

To reduce phishing risks, several measures can be implemented.

Technical Measures

- anti-phishing email filters
- multi-factor authentication (MFA)
- automated email analysis systems

User Awareness

- regular phishing awareness training
- phishing simulation exercises
- procedures for reporting suspicious emails

Conclusion.

Phishing remains one of the most common threats in cybersecurity. Systematic analysis of suspicious emails helps identify attacks and protect users from the theft of sensitive information.

The use of analysis tools combined with user awareness training represents an effective strategy to reduce the impact of phishing attacks